SCHOOL OF INFORMATION TECHNOLOGY

GAME DESIGN DOCUMENT (GDD)

VILL N' CHASE: CHASING TIME

VERSION NO. 7

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I. GAME OVERVIEW

1.1 High Concept Statement

Vill n' Chase: Chasing Time is not just any 2D platformer escape game, as the game combines silliness with a heart-pounding time limit and puzzles and challenges. Players play as "Villie the Villain," who must use their wits and skills to move around the Villain in Prison (VIP) Facility, avoiding traps, using items and mechanisms, and dealing with those who try to stop their escape.

1.2 Elevator Pitch

Players must put themselves in Villie's shoes who must escape Villain in Prison (VIP) Facility within a time limit. The player shall make use of items and VIP Facility's infrastructure and technology to get to the exit while avoiding obstacles and enemies. Players can choose from available maps and difficulties as they progress through the game. The Tutorial Map must be completed before the Main Map becomes available, while each difficulty must be completed before the next difficulty is unlocked. This allows players to learn the game mechanics and gradually face greater challenges while exploring the VIP Facility and solving puzzles.

1.3 Genre, Platform, and Target Audience

Genre and sub-genre	The game is a 2D platformer with puzzle-solving, hack and slash, and action-adventure elements.
Genre conventions the game keeps	Moving left, right, and jumping are to be expected as a 2D platformer game. As a puzzle-platformer, there will be doors to be unlocked with different solutions. Items that include weapons to be used on enemies make for its hack and slash aspect. When both puzzle-solving and hack and slash are paired along with exploration, we create an action-adventure game.

Genre conventions the game breaks	A time limit is sometimes part of puzzle games, but combining puzzle-solving with a time-limited escape, enemies, and traps gives Vill n' Chase its own challenge. Customizable difficulties also work with this to allow players to align with what suits them better in such a way that they can focus on what aspect, level of difficulty, or genre they can focus on more.
Target platform	Desktop PC; keyboard controls.
Target audience	Our target age rating would be 7 or older (PEGI 7). Players who enjoy short 2D platforming challenges, hazard avoidance, and time-based objectives; accessible to beginners because the tutorial introduces mechanics progressively.
Expected session length	Players may customize how long they want the game to last which can span from 3-10 minutes per in-game session.

1.4 Scope and Feasibility Statement

In scope for this term	Tutorial map; left/right movement; jump; hearts and timer HUD; Buff Police, Fast Boy, Witch and spikes; win state at the exit; lose state when time or hearts reach zero; health and speed potions
Out of scope (deliberately cut)	Multiplayer and online integration will not be included in the scope for this term.
Feasibility justification	Given the time limitations and abilities of the developers as of now, multiplayer and online integration will not be implemented this term.

II. FORMAL ELEMENTS

Players	The game is played by a single player who will play as Villie, with enemy movements programmed.
Objectives	• Escape the facility by reaching the exit before time runs out. • Escape with at least one heart remaining • There may be more objectives as difficulty increases
Rules	• The player can only carry one item at a time. • Locked doors can only be used when unlocked with a key or lever • The run ends when time or hearts reach zero.
Procedures	• Choose a map and a difficulty • Move and explore, find and use items, interact with mechanisms, avoid or deal with enemies and traps, find the exit • Unlock the exit and Escape
Resources	• One item capacity that can carry an item such as a Potion, Weapon, Repair Tool, or Key • Hearts and remaining time. Starting values depend on difficulty: ○ Easy: 10 hearts/600 seconds; ○ Medium: 8 hearts/420 seconds; ○ Hard: 6 hearts/300 seconds.
Conflict	• Fast Boy, Buff Police, and spikes can reduce hearts on contact • Witch can polymorph Villie for 4 seconds • The player should reach the exit before time runs out.
Boundaries	The game takes place within the facility selected by the player. The game is played on a desktop using keyboard controls.
Outcome	Win: • Villie reaches the exit while at least one heart remains and time is above zero. Lose: • Hearts reach zero, or the timer reaches zero.

III. MDA AND PLAYER EXPERIENCE

3.1 Mechanics, Dynamics, and Aesthetics

Mechanic	Dynamic it is expected to produce	Aesthetic / kind of fun it serves
Time limit	The player balances speed against safe movement as the countdown decreases; harder settings allow less time.	Challenge
Moving and jumping on platforms	The player moves around the facility with simple left and right movement along with jumping to move to platforms and avoid spikes and Fast Boy.	Discovery/Challenge
Buff Police collision	Because he is tall and cannot be jumped over, the player must avoid contact and use available space or another route.	Challenge
Fast Boy collision	His speed creates a timing challenge, but his small size allows the player to jump over him.	Challenge
Witch polymorphs	When Villie gets too close, Villie transforms into a hideous creature for 4 seconds; the effect has a 4-second cooldown.	Challenge

Mechanic	Dynamic it is expected to produce	Aesthetic / kind of fun it serves
Carry one limit at a time	Players can carry an item until such a point that another item must be used. Afterwards, they can pick up the dropped item again. For example, dropping a key to pick up a weapon to knock out an enemy near the door.	Challenge
Enter doors	Players enter doors to go from one point of the map to the other. This can be used as a way to time enemy movement by entering a door and going back. Doors also allow access to areas that have items and paths.	Discovery
Unlock locked doors and the exit with keys or mechanisms	Players explore the facility in search of keys or mechanisms that would open doors and the exit. With a time and item limit and the presence of traps and enemies, players may need to think about the necessity of unlocking each door.	Discovery/Challenge

Mechanic	Dynamic it is expected to produce	Aesthetic / kind of fun it serves
Knockout enemy with a weapon	Players can knock out enemies to safely walk around a platform. Knocking out an enemy may be easier than finding ways to avoid them.	Fantasy/Challenge
Use potion	Players can use potions like Health potions to restore some heart, increasing their chances of survival. They may use Immunity potions when they know that an enemy or trap with a debuff is near, and facing them head-on with an Immunity potion would be faster than avoiding them. A Boost potion gives them the speed they need to clutch their way to the exit.	Challenge
Use mechanisms	Players can use mechanisms in the facility like levers that open doors, allowing them more access to areas.	Discovery/Challenge

3.2 Targeted Kinds of Fun

Included

- Fantasy - Playing as Villie puts players in the boots of a supervillain that attempts to escape against all odds. Players make use of technology in the facility or items that don't exist in the real world.

- Challenge - The time limit, enemies, traps, and code puzzle require players to make quick decisions to overcome obstacles
- Discovery - Players must explore the facility/ies to find useful items to overcome obstacles and open the locked door/exit

Excluded

- Fellowship - As the game is a single-player offline game, players will not be able to interact or play with with others in the game

3.3 Player Types and Motivation

The game is mainly designed for "explorers" and "achievers". Explorers are supported through searching the facility for items, code, and mechanism interaction. Achievers are supported through completing objectives such as finding the necessary mechanisms and items, overcoming obstacles, and escaping before the time runs out.

3.4 Player Experience (PX) Goals

PX goal (what the player should feel, and when)	Design feature intended to produce it	How you would observe it in a playtest
The player should feel a sense of urgency when the timer goes lower	• Time limit • Timer for the in-game session	Observe whether players make quick decisions as the time decreases
The player should feel cautious when approaching traps and enemies	• Visible Buff Police, Fast Boy, Witch, and stationary spikes; distinct sizes and behaviors.	Observe whether players attempt different strategies to deal with traps and enemies

PX goal (what the player should feel, and when)	Design feature intended to produce it	How you would observe it in a playtest
The player should feel rewarded for using different items and mechanisms to overcome obstacles	• Weapons for dealing with enemies • Potions for the player • Mechanisms for unlocking doors, disabling traps, or moving around • Doors for moving around the facility	Observe whether players use items and mechanisms to solve problems and progress
The player should feel curious and motivated in trying to find and unlock the exit	• Code displayed in different areas per in-game session • Mechanism to unlock the exit • Multiple possible exits, but only one is usable per in-game session	Observe whether players actively check different areas for the code, the specific mechanism, and the exit

IV. CORE GAMEPLAY LOOP

4.1 Core Loop Diagram

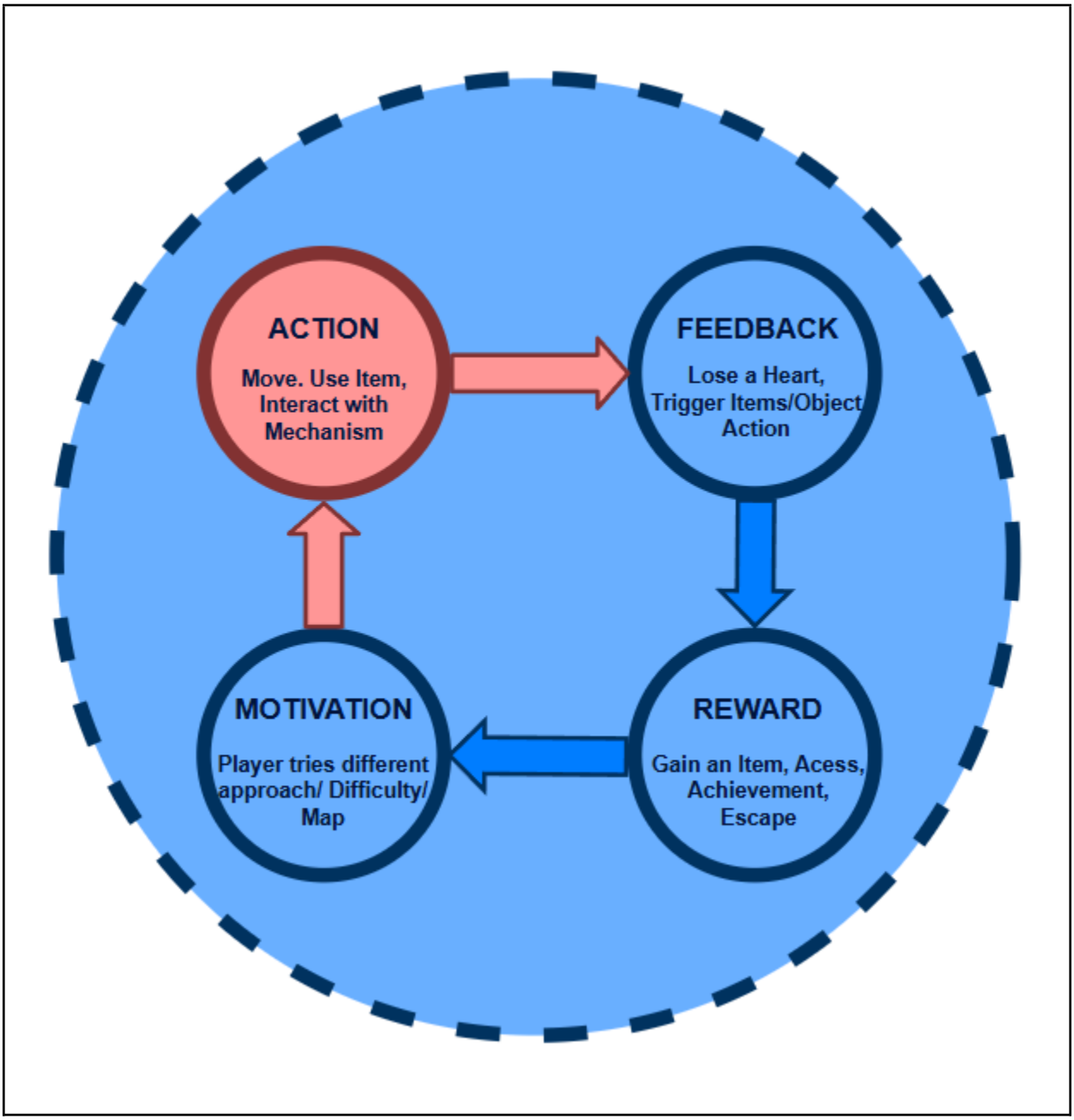


Figure 1. Core Loop Diagram

4.2 Loop Walkthrough

The core gameplay loop begins with Action, where the player performs an action based on the current situation in the facility. This may include moving, jumping, avoiding a trap, dealing with an enemy, using an item, or interacting with a mechanism.

The player action then produces Feedback from the game. Depending on the action, the player may lose a heart when a player bumps to an enemy or triggers a trap, or an item or interactive mechanism may trigger its intended action or effect when used.

The player then receives a Reward or Progress from successfully responding to the situation. This may include gaining an item, gaining access to a previously blocked area, gaining an achievement, or escaping the current facility and advancing to the next level.

The resulting progress provides Motivation for the player to continue. The player may decide to try a different approach, explore another route, play a different map, or try another level of difficulty which results in restarting the core gameplay loop.

One complete pass through the core gameplay loop is expected to take approximately 10-30 seconds, depending on the player's action, and the player's decision-making speed.

4.3 Macro and Micro Loops

Micro loop	Move around the facility → avoid enemies and traps → find or use item/s → interact with mechanisms → continue toward the exit.
Macro loop	Choose a map and difficulty → start the game → explore the facility → deal with enemies, traps, and obstacles → manage items and time → find and reach the exit → escape or lose.
Relationship between them	The player repeats the micro loop while exploring the facility. Completing these small actions helps the player move through the facility, overcome obstacles, and get closer to the exit. Repeating the micro loop eventually leads to completing the macro loop by reaching the exit and escaping.

4.4 In-Game and Out-Game Actions

In-game actions	Move left and right, jump, enter doors, pick up and use items, interact with mechanisms in the facility, avoid enemies and traps, use weapons on enemies, and find the exit.
Out-game actions	Choose a map, choose or customize the difficulty, start a game, or play again after the game ends.

V. LEVEL DESIGN

5.1 Level Layout Pattern

The game follows a linear layout where players have to complete a time-bound play. Once the escape starts, the timer starts and ends with little to no stop when the time runs out, the player loses hearts, or escapes.

5.2 Annotated Level Map

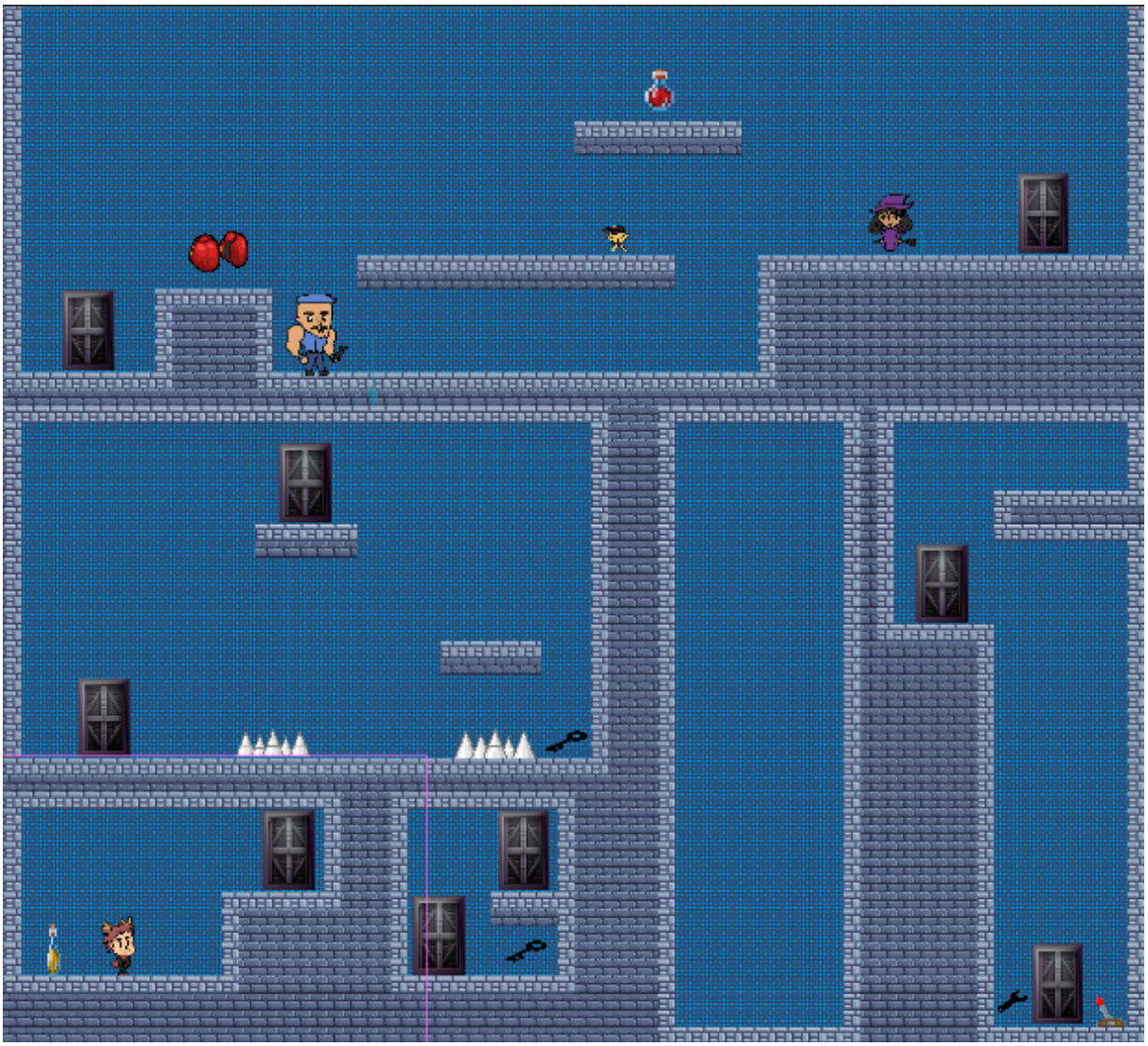


Figure 2. Full tutorial map



Figure 3. Player's view on game start

1. **Player start:** where Villie spawns
2. **Doors:** mechanism that transports Villie to a different door
3. **Keys:** used to unlock locked doors
4. **Levers:** special mechanism used to unlock doors or the exit
5. **Tool:** used to fix broken levers
6. **Exit:** the objective at the end of the route.
7. **Platforms:** safe ground and jumping surfaces.
8. **Spikes:** stationary traps dealing 1 heart of damage; jump over them.
9. **Buff Police:** slow and tall; deals 3 hearts of damage and cannot be jumped over.
10. **Fast Boy:** small and fast; deals 2 hearts and can be jumped over.
11. **Witch:** stationary, same approximate height as Villie, jumpable; transforms Villie into a hideous creature for 4 seconds when too close, with a 4-second cooldown.

The map is a set of stacked platforms linked by doors and jump springs. Each area in this map shows a new mechanic starting from potions and doors, then unlocking a locked door with a key, then avoiding spikes, then avoiding different kinds of enemies or straight up dealing with them, and finally unlocking the exit.

5.3 Pacing and Difficulty Curve

The game uses a rising-tension curve with short relief moments, so players feel pressure without being overwhelmed. This follows the idea that challenges should grow with the player's skill to keep them in a state of flow. Because each session lasts 3 to 10 minutes, pacing is tied to the timer and divided into three phases.

1. **Opening (low tension):** picks up the first item, jumps across platforms, and moves through doors.
2. **Middle (rising tension):** the player juggles the one-item limit, enemies, and locked doors while hunting for the key or code. Enemies and spikes begin appearing as well.
3. **Final rush (peak tension):** the timer is low, and the player must reach the exit.

The game uses a rising-tension curve with short relief moments, so players feel pressure without being overwhelmed. This follows the idea that challenges should grow with the player's skill to keep them in a state of flow. Because each session lasts 3 to 10 minutes, pacing is tied to the timer and divided into three phases.Difficulty settings change the curve by adjusting the timer length, number of enemies, trap density, and additional objectives. Easy lowers the peak of the final rush, while Hard raises it and shortens the opening phase.

5.4 Teaching Mechanics Through Level Sequencing

Mechanic to be taught	Where it is introduced	How the level teaches it
Timer and hearts	HUD visible from run start	The HUD updates as time passes and damage is received, showing the cost of unsafe choices.
Movement (Left, Right, Jump)	Near spawn	High elevated platforms force jumping with no penalty

Mechanic to be taught	Where it is introduced	How the level teaches it
Doors	Near spawn	An unlocked door stands alone while another door of the same color is seen past the wall, hinting that by interacting with the door, they can go to the next area.
Speed potion	Near spawn	Floats near the player at the start of the game, attracting the player to pick it up and use it.
Keys and locked doors	First locked door, key visible nearby	Color match shows which key fits
Pick up and one-item limit	When they attempt to carry another item while they're already carrying one; the earliest point this can happen is if they bring the speed potion to the second area and they attempt to grab the key	Player must drop one to take the other
Spikes	After jumping and locked doors were taught	Spiked with enough space
Fast Boy	After spikes were taught; on its own in one long platform	Its small size and movement show that jumping over it is a viable response.

Mechanic to be taught	Where it is introduced	How the level teaches it
Buff Police	After spikes were taught; on its own below the fast boy's platform	Its tall body and slow movement show that it cannot be jumped over; the player must avoid contact.
Healing potion	After Fast Boy and Buff Police were shown	After encountering different enemies, a potion will be on a platform which they can take to heal damage taken
Witch (if implemented)	After spikes were taught; without any other enemy on its platform	The stationary enemy can be jumped over, but proximity triggers a 4-second transformation; the effect has a 4-second cooldown.
Exit mechanism	Final area	Combines everything learned earlier

VI. SYSTEMS AND BALANCE

6.1 Resource Economy

Resource	Start value	Source	Sink
Hearts	• Easy - 10 • Normal - 8 • Hard - 6	• Set at the start of a run according to selected difficulty. • A Health potion restores 2 hearts, up to the starting maximum.	Hearts are lost when Villie collides with an enemy or spikes: • Buff Police - 3 • Fast Boy - 2 • Spikes - 1
Time	• Easy - 600 s	Set at run start according to	Counts down once the run begins.

	• Normal - 420 s • Hard - 300 s	selected difficulty.	
Item slot	1 empty slot	Set at run start	Picking up an item causes the slot to be occupied and other items cannot be picked up
Keys	The player has no key held	Fixed map locations; There is one key per pair of locked doors	Each key opens one matching lock and is consumed
Gloves	The player has no glove held	Fixed map locations	3 uses. Each use knocks out nearby enemies for 5 s. Removed after the third use.
Potions	The player has no potion held	Fixed map locations	Single use. Health: +2 hearts. Boost: +50% move speed for 8 s.
Repair tools	The player has no repair tool held	Fixed map locations	Consumed when used on a broken lever.

6.2 Progression Curve

The progression curve below shows the growth of difficulty as the player gets closer to the exit.

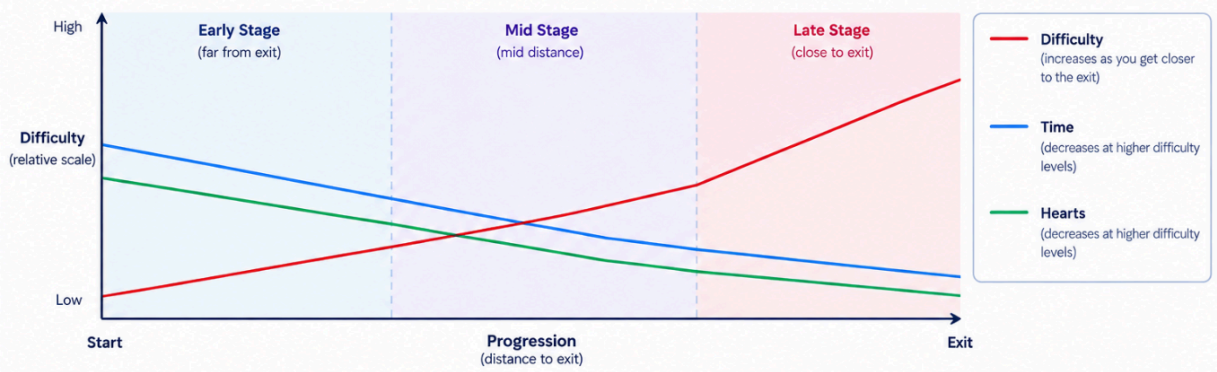


Figure 4. Progression Curve

As shown in the figure, difficulty increases while time decreases as they get closer to the exit. Difficulty here may pertain to the number of enemies, spikes, locked doors, broken levers, or the combination of obstacles. Players will make use of basic mechanics at the start with little difficulty. They may struggle as the game progresses where more skill is required and the remaining time is lower. They may also lose hearts along the way due to the increasing number of enemies and spikes.

6.3 Symmetry, Asymmetry, and Fairness

Vill n' Chase is an asymmetric single-player game. The player controls Villie, who has movement, items, and mechanisms. Enemies follow programmed patrols and abilities and have no goals of their own, so the player and enemies do not share controls or objectives. Still, fairness is maintained between the player and the systems opposing them.

When it comes to enemies, they appear at least one screen before they can reach Villie. There are also different types of enemies that have advantages and disadvantages. The Buff Police is big but slow, the Fast Boy is small enough to jump over but moves fast, and the Witch does not move, but can cast a debuff for 4 seconds when within her casting range. Also, Villie is intended to be invulnerable for 1 second after receiving a hit to avoid

continuously damaging Villie in a short amount of time or before the player can get Villie out of a spike.

Regardless of difficulty, players must be able to accomplish the game. As such, these have been tested, and the level of skill needed to finish a game on a given difficulty is appropriate. Changes to time and hearts are tuned per difficulty while still ensuring that the player can get Villie to escape in any difficulty.

6.4 Randomness and Probability

The game has no element of randomness. Damage values, locations of items, and item effects are fixed. Random exit and item placements may cause unfairness as players may waste time while others may find themselves lucky. Lastly, not implementing randomness keeps the game simple and achievable enough given the time constraints.

VI. NARRATIVE DESIGN

Not Applicable. The game does not focus on providing narrative content to the player as it focuses more on fantasy, challenge, and submission among the kinds of fun. For every level, the game immediately starts with Willie at the start of the prison break with the timer about to run. The player is meant to focus on getting Villie to escape in time from the beginning, rather than learning more about a story.

VII. TECHNICAL DESIGN

8.1 2D, 3D, or Hybrid Decision

Vill N' Chase will be developed as a 2D game. The decision was made given the time constraints, the size of the team, and the current skill level of the developers. With these in mind, a simple platformer game will become the deliverable. For readability, players will be able to differentiate walkable platforms, enemies, spikes, and mechanisms from the background. Movement of the player and the enemies is more predictable and easier to understand as they move on a flat surface. Enemies can only walk or run left to right and vice versa, while the Player can also jump in the same

directions. The distance of a player's jump is simpler to implement and understand in 2D.

8.2 Camera System

Camera type	Side-scroller with the camera centered on the player
What the camera must always keep visible	Villie, the current platform, nearby hazards, the exit when possible, and the hearts/time HUD.

Readability risks	- *Limited Vision:* Players should only be able to see the area surrounding Villie. This means that players may have trouble navigating, looking for items, and watching out for spikes and enemies. While this is part of the challenge, fairness between the player and the game's system was considered in level design. As much as possible, players will be able to spot a buff police or fast boy before they can get away from them. They should also be able to spot a witch before suffering its effects. - Differentiation: Players should be able to differentiate platforms vs. background so they know where it is safe to land. Doors and keys shall be different from one another as well. They will come in different colors, with the doors having matching colors with their corresponding key/s. Players should also be able to tell the state of a lever (up, down, broken). Aside from showing the lever's position, there will be different colors per state. Finally, to make differentiation easier, the background and the rest of the elements in the game will have some differences in lighting and art.

8.3 Control Scheme and Input Mapping

Player action	Primary input	Alternate input	Rationale
Walk Left	A	[<-] Left Arrow	Move left.
Walk Right	D	[->] Right Arrow	Move right.
Jump	W or SPACE BAR	Up Arrow[space]	Jump over gaps, spikes, and jumpable enemies.

Player action	Primary input	Alternate input	Rationale
• Pick up item • Use item • Open door • Interact with lever	E		Interact with items and mechanisms
Drop item	Q		Allow Villie to carry another item

8.4 Godot Scene Tree Plan

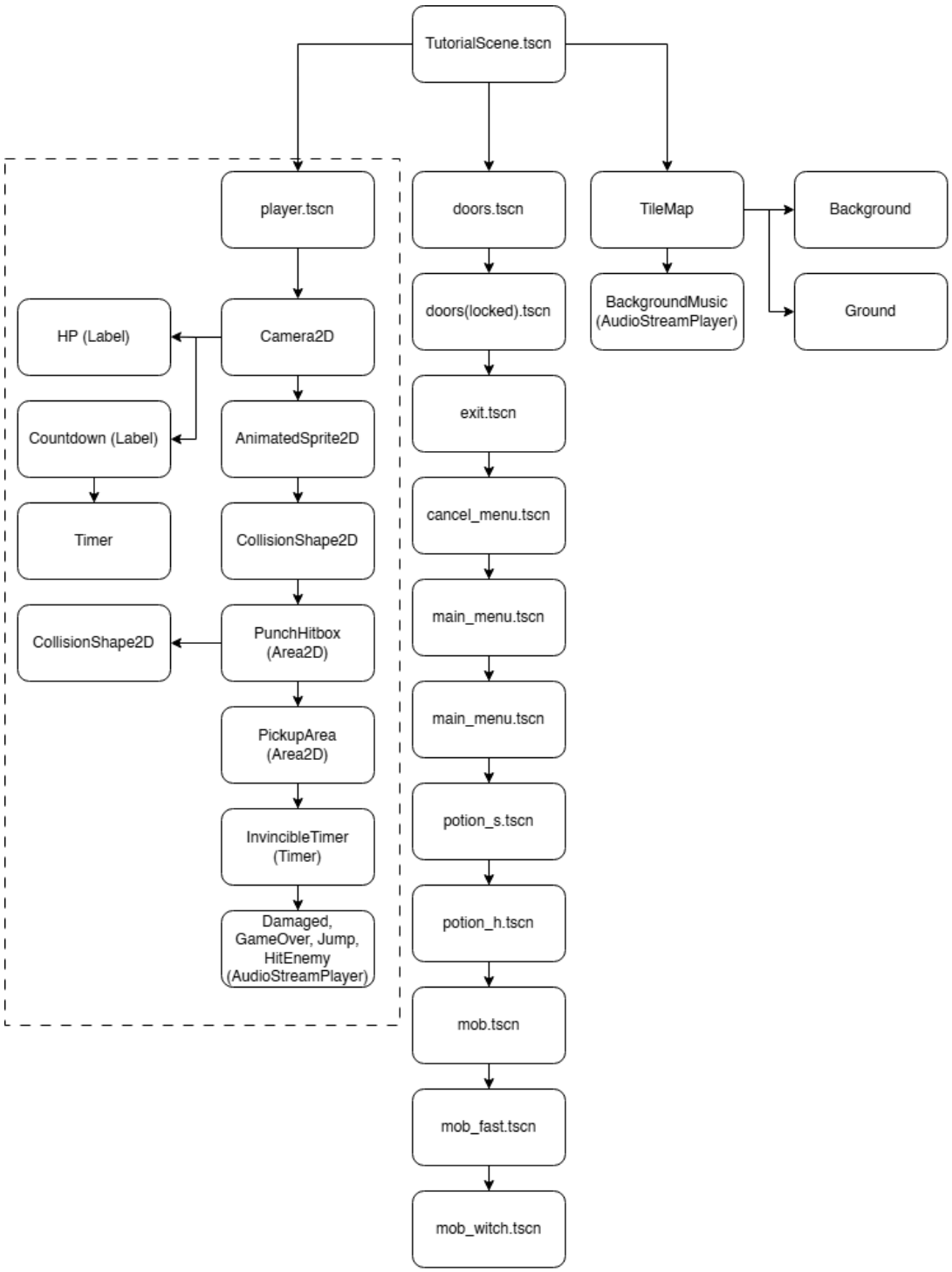


Figure 5. Scene Tree of TutorialScene

The figure above shows the Scene Tree of TutorialScene where the Player scene is also expanded as it is another integral scene.

8.5 Signals and Event Wiring

Signal	Emitted by	Connected to	What happens as a result
body_entered	Door 1	_on_body_entered()	Detects when the player enters the first door's area.
body_exited	Door 1	_on_body_exited()	Detects when the player leaves the first door's area.
body_entered	Door 2	_on_body_entered()	Detects when the player enters the second door's area.
body_exited	Door 2	_on_body_exited()	Detects when the player leaves the second door's area.
body_entered	Locked Door 1	_on_body_entered()	Detects when the player approaches the locked door.
body_exited	Locked Door 1	_on_body_exited()	Detects when the player leaves the locked door's area.
body_entered	Locked Door 2	_on_body_entered()	Detects when the player approaches the second locked door.
body_exited	Locked Door 2	_on_body_exited()	Detects when the player leaves the second locked door's area.

Signal	Emitted by	Connected to	What happens as a result
body_entered	Key	_on_body_entered()	Detects the player and allows the key to be picked up.
body_exited	Key	_on_body_exited()	Detects when the player moves away from the key.
body_entered	Gloves	_on_body_entered()	Detects the player and allows the gloves to be picked up.
body_exited	Gloves	_on_body_exited()	Detects when the player moves away from the gloves.
body_entered	Healing Potion	_on_body_entered()	Detects the player entering the potion's pickup area.
body_exited	Healing Potion	_on_body_exited()	Detects when the player leaves the potion's pickup area.
body_entered	Speed Potion	_on_body_entered()	Detects the player entering the potion's pickup area.
body_exited	Speed Potion	_on_body_exited()	Detects when the player leaves the potion's pickup area.

Signal	Emitted by	Connected to	What happens as a result
body_entered	Repair Kit	_on_body_entered()	Detects the player entering the repair kit's pickup area.
body_exited	Repair Kit	_on_body_exited()	Detects when the player leaves the repair kit's pickup area.
body_entered	Lever	_on_body_entered()	Detects when the player approaches the lever
body_exited	Lever	_on_body_exited()	Detects when the player leaves the lever's area.
body_entered	Broken Lever	_on_body_entered()	Detects when the player approaches the broken lever.
body_exited	Broken Lever	_on_body_exited()	Detects when the player leaves the broken lever's area.
body_entered	Exit Door	_on_body_entered()	Detects when the player enters the exit door's area.
body_exited	Exit Door	_on_body_exited()	Detects when the player leaves the exit door's area.
body_entered	Spike	_on_body_entered()	Detects when the player touches the spikes and deals damage.

Signal	Emitted by	Connected to	What happens as a result
body_entered	Fat Spike	_on_body_entered()	Detects when the player touches the larger spike hazard and deals damage.
body_entered	Fast Boy Hitbox	_on_hitbox_body_entered()	Detects player collision with the faster enemy and deals damage.
timeout	Buff Police Knockout Timer	_on_ko_timer_timeout()	Restores the enemy's health and reactivates its hitbox after being knocked out.
timeout	Fast Boy Knockout Timer	_on_ko_timer_timeout()	Restores the fast enemy's health and reactivates its hitbox after being knocked out.
timeout	Witch Cooldown Timer	_on_cooldown_timeout()	Allows the witch to use its petrification ability again.

Signal	Emitted by	Connected to	What happens as a result
timeout	Witch Petrification Timer	_on_p_time_timeout()	Ends the petrification effect and restores the player's movement.
timeout	Witch Knockout Timer	_on_ko_timer_timeout()	Restarts the witch’s initial status.
timeout	Player Countdown Timer	_on_timer_timeout()	Triggers the player's death when the countdown reaches zero.
pressed	Restart Button (Death Menu)	_on_restart_pressed()	Reloads the current game scene after the player loses.
pressed	Restart Button (Completion Menu)	_on_restart_pressed()	Reloads the current game scene after completing the level.
pressed	Continue Button	_on_continue_pressed()	Resumes the game after closing the menu.
pressed	Restart Button (Pause Menu)	_on_restart_pressed()	Restarts the current game scene from the pause menu.

Signal	Emitted by	Connected to	What happens as a result
pressed	Start Button	_on_start_pressed()	Loads the main gameplay level when the player starts the game.

XII. REFERENCES AND ASSET ATTRIBUTION

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12.2 Asset Attribution

Asset	Type	Source	Licence and permitted use
Villie Sprites	Image	Isaac A. Bagsangi	Original
Witch Sprites	Image	Isaac A. Bagsangi	Original
Buff Police Sprites	Image	Isaac A. Bagsangi	Original
Fast Boy Sprites	Image	Isaac A. Bagsangi	Original

Other assets were taken from the internet and currently serve as placeholders and may be replaced later on. Attribution will be provided.